Daily Logic Drill #6

Sheet by David Akinyele-Aje

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Mathematical Induction · Strong Induction · Base Case Traps · Misapplied Induction.

Section A

1. A sequence is defined by $u_1 = 1$ and $u_{n+1} = 2u_n + 1$. A student claims $u_n = 2^n - 1$ for all $n \geq 1$. For which value of n does the base case verify this claim?

Ans: 1

2. To prove $P(n)$ for all integers $n \geq 4$ by induction, what is the base-case integer?

Ans: 4

3. If $P(k) \implies P(k+1)$ is proved for all $k \geq 1$, but no base case is ever established, has $P(n)$ been proved for any n?

Ans: False.

4. A sequence satisfies $u_{n+1} = u_n + 2n + 1$ with $u_1 = 1$. What is the closed-form expression for u_n?

Ans: n^2

5. True or false: "Mathematical induction can be used to prove statements about every real number $x \geq 1$."

Ans: False.

6. To prove $2^n > n^2$ for all $n \geq 5$ by induction, what is the smallest integer n for which the base case must be verified?

Ans: 5

7. What is the negation of the inductive-step statement "For all $k \geq 1$, $P(k) \implies P(k+1)$"?

Ans: $\exists k \geq 1$ such that $P(k) \wedge \neg P(k+1)$

8. True or false: "If $P(1)$ is true and $P(k) \implies P(k+2)$ holds for all $k \geq 1$, then $P(n)$ is true for every positive integer n."

Ans: False.

9. What is the smallest positive integer n for which $n! > 3^n$?

Ans: 7

10. A recurrence satisfies $u_{n+2} = 3u_{n+1} - 2u_n$ with initial terms $u_1 = 1$ and $u_2 = 3$. What is the minimum number of base cases required to anchor an induction for all integers $n \geq 1$?

Ans: 2

Section B

1. Express $\sum_{i=1}^{k+1} i^3$ in terms of $\sum_{i=1}^{k} i^3$.

Ans: $\sum_{i=1}^{k+1} i^3 = \sum_{i=1}^{k} i^3 + (k+1)^3$

2. Assuming $\sum_{i=1}^{k} i = \frac{k(k+1)}{2}$, add $(k+1)$ and simplify to show $\sum_{i=1}^{k+1} i = \frac{(k+1)(k+2)}{2}$.

Ans: $\frac{k(k+1)}{2} + (k+1) = \frac{k(k+1)+2(k+1)}{2} = \frac{(k+1)(k+2)}{2}$

3. Prove by induction: $\sum_{i=1}^{n}(2i-1) = n^2$ for all positive integers n.

Ans: Proof by induction

4. Assuming $4^{2k} - 1$ is divisible by 15, express $4^{2(k+1)} - 1$ in a form that shows it is also divisible by 15.

Ans: $4^{2(k+1)} - 1 = 16(4^{2k} - 1) + 15$

5. Prove by induction that $5^n - 1$ is divisible by 4 for all positive integers n.

Ans: Proof by induction

6. Assuming $k! > 2^k$ for some $k \geq 4$, prove $(k+1)! > 2^{k+1}$.

Ans: Proof

7. Prove by induction: $2^n \geq n+1$ for all non-negative integers n.

Ans: Proof by induction

8. Assuming $\sum_{i=1}^{k} \frac{1}{i(i+1)} = \frac{k}{k+1}$, show that $\sum_{i=1}^{k+1} \frac{1}{i(i+1)} = \frac{k+1}{k+2}$.

Ans: $\frac{k}{k+1} + \frac{1}{(k+1)(k+2)} = \frac{k(k+2)+1}{(k+1)(k+2)} = \frac{(k+1)^2}{(k+1)(k+2)} = \frac{k+1}{k+2}$

9. Prove by induction that the sum of the interior angles of a convex n-gon is $(n-2) \times 180^\circ$ for all $n \geq 3$.

Ans: Proof by induction

10. A student wants to prove $P(n)$ for all $n \geq 1$ using base cases $P(1)$ and $P(2)$, with inductive step "$P(k)$ and $P(k+1)$ together imply $P(k+2)$" for $k \geq 1$. Explain why establishing *both* $P(1)$ and $P(2)$ (not just $P(1)$) is necessary here.

Ans: The inductive step requires two consecutive prior cases to fire; without $P(2)$ independently established, the chain can never produce $P(2)$ or anything depending on it.

Section C

1. A flawed "proof" that everyone in any group shares a favourite integer.

Ans: B

2. Evaluate an inductive step for $n^3 - n$ divisible by 6 that appeals to "the general theory" instead of doing the algebra.

Ans: B

3. Evaluate a base-case dispute for $n^2 > 3n + 4$.

Ans: B

4. Evaluate a weak-induction attempt to prove every $n \geq 2$ is a product of primes.

Ans: B

5. Evaluate whether a failed $P(1)$ means no base case exists for $\frac{1}{n+1} + \cdots + \frac{1}{2n} > \frac{1}{2}$.

Ans: B

6. Evaluate the "3p/5p postage" strong-induction step.

Ans: A

7. Evaluate a finite-case "proof" of the alternating Fibonacci sum identity.

Ans: B

8. Which statement precisely captures the inductive step?

Ans: C

Section D

1. $t = x + \frac{1}{x}$ integer > 2; define $t_n = x^n + \frac{1}{x^n}$. *(after BMO1 2015 Q4)*

Ans: (a) $t_{n+1} = t \cdot t_n - t_{n-1}$. (b) Proof by strong induction below.

2. Prove every positive integer is a sum of distinct charming integers (2 or $3^i 5^j$). *(after BMO1 2016 Q6)*

Ans: Proof by strong induction below, via the key gap lemma.

3. Prove Bernoulli's Inequality: $(1+x)^n \geq 1+nx$ for $x > -1$, n a positive integer.

Ans: Proof by induction below; the condition $x > -1$ is required when multiplying the inequality by $(1+x)$.

4. $a_1 = 2, a_{n+1} = a_n^2 - a_n + 1$. Prove $a_1, \ldots, a_n$ pairwise coprime.

Ans: Proof by induction below (via the lemma $a_n - 1 = \prod_{i=1}^{n-1} a_i$).

5. Prove $\sum_{k=1}^{n} \frac{1}{k^2} < 2 - \frac{1}{n}$ for all integers $n \geq 2$.

Ans: Proof by induction below.